WEEK 6: ASSIGNMENT

1. What will be the flux swing in case of forward converter

A. $-\Phi_{max}$ to Φ_{max}

B. 0 to Φ_{max}

C. 0 to $0.5\Phi_{max}$

D. $-0.5\Phi_{max}$ to $0.5\Phi_{max}$

Answer: B

2. If the q-axis component of current is 10A and the d-axis component of current is 8A in case of dq axis based control of three-phase AC-DC converter, then what will be the power factor of the drawn current. If the d-axis is aligned along the supply voltage space vector. Note: The answer should be rounded up to 3 decimal places.

Answer: 0.600 to 0.650

3. If the q-axis component of supply current is double of the d-axis component of supply current in the case of dq-based control of a three-phase AC-DC converter, then what will be the power factor of the drawn current, if the q-axis is aligned along the supply voltage space vector? Note: The answer should be rounded up to 3 decimal places.

Answer: 0.870 to 0.920

4. In three-phase AC-DC converter control, aligning supply current vector with the q-axis ensures:

A. unity power factor current drawn
B. zero power factor current drawn
C. power flow control from inverter to grid
D. Stable operation

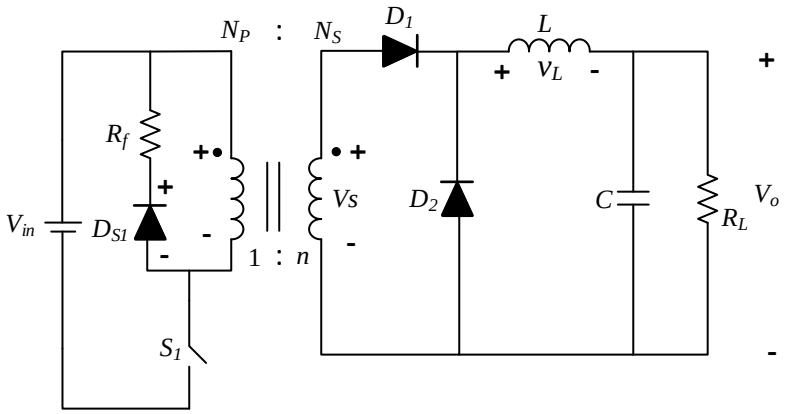
Answer: B

5. Which of following statement is true in case of d-q based three-phase AC-DC converter control

A. d, q components are independent of each other
B. d, q components are dependent of each other
C. α, β components are independent of each other
D. α, β components are dependent of each other

Answer: A, C

6. In case of forward isolated DC-DC converter what will be the maximum voltage stress across the diode D_{S1}? Note: Symbols have obvious meaning



A. V_{in}
B. $2V_{in}$
C. V_{in} plus the voltage drop across the resistance used for reset the core flux
D. $V_{in}/2$

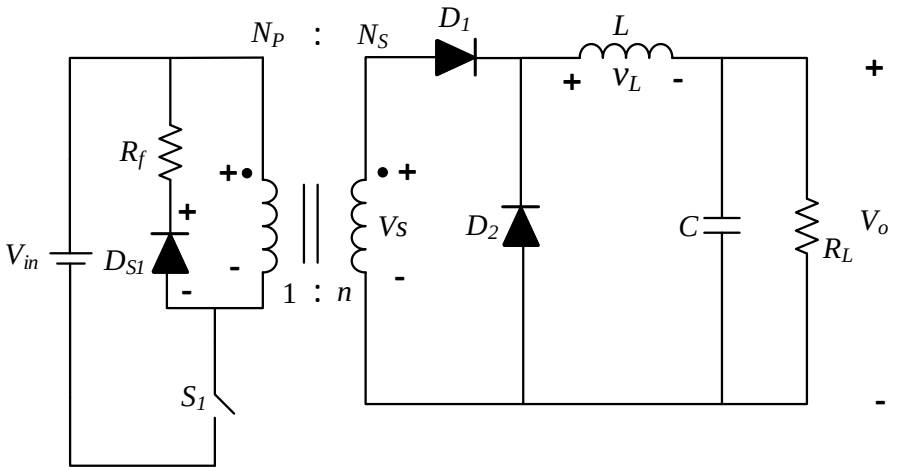
Answer: A

7. A dq transformation applied to balanced 3-phase voltages having frequency of 60Hz gives

A. α, β components which are constant
B. α, β components which have 60Hz sinusoidal variation
C. d, q components which are constant
D. d, q components which have 60Hz sinusoidal variation

Answer: B, C

8. The forward converter shown below has a V_{in} = 400 V and a transformer having a turns ratio (N_p:N_s) of 5:2.



If the switch S_1 is turns-ON for 10 µsec in a switching cycle and turns-OFF for 40 µsec, then what will be the output voltage. Note: The answer should be rounded up to 1 decimal places.

Answer: 30 to 35

9. For a three-phase AC-DC converter using 12mH having 10mΩ internal resistance, conventional PI controller is used for inner loop current control with K_{pc} and K_{ic} as controller gains. If the controller is selected to cancel the pole introduce dues to inductance and resistance. Then, what will be the value of K_{pc}/K_{ic}? Note: Symbols have obvious meaning

A. 1.2
B. 0.834
C. 0.1
D. 1
Answer: A

10. Which of the following relationship is true for the time constant $T_s = L/R_s$ and T_d?. Note: Symbols have obvious meaning

A. $T_s < T_d$
B. $T_s > T_d$
C. $T_s = T_d$

Answer: B

Answer Key:

1. B	2. 0.60 to 0.65
3. 0.87 to 0.92	4. B
5. A, C	6.A
7. B, C	8. 30 to 35
9. A	10. B